

5.5 The normal distribution

Probability and Statistics 1 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Normal distributions are symmetric with mean μ and standard deviation σ ; standardise $z=(x-\mu)/\sigma$ before using tables. A continuous distribution has $P(X=a)=0$. The normal approximation to a discrete model needs suitable conditions and continuity correction where applicable.

Worked example: follow the reasoning

If $X \sim N(100, 15^2)$, $x=115$ has $z=1$; $P(X \leq 115) \approx 0.8413$.

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 For $X \sim N(50, 10^2)$, express $P(40 < X < 60)$ in standard normal form and estimate it. [3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

z bounds -1 and $+1$ [1]; probability $\Phi(1) - \Phi(-1)$ [1] ≈ 0.6827 [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.